YUYU LIN

PhD student
Carnegie Mellon University, Human-Computer Interaction Institute
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RESEARCH INTERESTS

Personal Exoskeleton, Computational Fabrication, Dynamic Structures

My current research focuses on the **design, customization and fabrication of rehabilitative and assistive wearables.** I broadly explore ***active materials*** and ***passive structures*** with adaptive properties that respond to the user's dynamic requirements and body movements. I propose ***simulation and customization pipelines*** to enable end users to personalize their wearables.

EDUCATION

08/2022 – 08/2027 (expected) **Carnegie Mellon University,** Pittsburgh, USA
PhD. in Human Computer Interaction
Advisor: Prof. Alexandra Ion

09/2021 – 08/2022 **Stanford University,** Stanford, USA
M.S. in Computer Science (Specialization: Artificial Intelligence)
Research with Prof. Hari Subramonyam, Steven M. Drucker (Microsoft), Prof. Dakuo Wang, Elyse Chase, Eric Gonzalez

09/2018 – 03/2021 **Zhejiang University,** Hangzhou, China
M.E. in Design Engineering
Advisor: Prof. Fangtian Ying

09/2014 – 06/2018 **Zhejiang University,** Hangzhou, China
B.E. in Industrial Design

07/2017 **Stellenbosch University**
Stellenbosch, South Africa
Summer Program

PUBLICATIONS

Yuyu Lin, Yujia Liu, Emma Kim, and Alexandra Ion. "Towards Accessible Mobility Support: User-Centered Design of a Passive, Multi-Functional, Low-Cost Knee Exoskeleton". In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26). 2026.

Yuyu Lin, Dian Zhu, Anoushka Naidu, Kenneth Yu, Deon Harper, Eni Halilaj, Douglas Weber, Deborah Ellen Kenney, Adam J. Popchak, Mark Baratz, and Alexandra Ion. "Personalized Bistable Orthoses for Rehabilitation of Finger Joints." In The 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25). 2025.

Best Demo Honorable Mentioned Award

Yuyu Lin, Hatice Gokcen Guner, Jianzhe Gu, Sonia Prashant, and Alexandra Ion. "Wearable Material Properties: Passive Wearable Microstructures as Adaptable

Interfaces for the Physical Environment." In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). 2025.

Yuyu Lin, Jesse T. Gonzalez, Zhitong Cui, Yash Rajeev Banka, and Alexandra Ion. "ConeAct: A Multistable Actuator for Dynamic Materials." In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, pp. 1-16. 2024.

Violet Yinuo Han, Abena Boadi-Agyemang, **Yuyu Lin**, David Lindlbauer, and Alexandra Ion. "Parametric Haptics: Versatile Geometry-based Tactile Feedback Devices." In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 2023.

Jianzhe Gu, **Yuyu Lin**, Qiang Cui, Xiaoqian Li, Jiaji Li, Lingyun Sun, Cheng Yao, Fangtian Ying, Guanyun Wang, and Lining Yao. "PneuMesh: Pneumatic-driven Truss-based Shape Changing System." In *CHI Conference on Human Factors in Computing Systems*, pp. 1-12. 2022.

Yuyu Lin, Jiahao Guo, Yang Chen, Cheng Yao, and Fangtian Ying. "It is your turn: collaborative ideation with a co-creative robot through sketch." In *Proceedings of the 2020 CHI conference on human factors in computing systems*, pp. 1-14. 2020.

Guo Jiahao, **Yuyu Lin**, Hongyu Yang, Junwu Wang, Shuo Li, Enmao Liu, Cheng Yao, and Fangtian Ying. "Comparing the tangible tutorial system and the human teacher in intangible cultural heritage education." In *Proceedings of the 2020 ACM Designing Interactive Systems Conference*, pp. 895-907. 2020.

Liu Lijuan, Cheng Yao, **Yuyu Lin**, Yang Chen, Ziyu Liu, and Fangtian Ying. "AquaBot: An interactive system for digital water play." In *International Conference on Human-Computer Interaction*, pp. 16-27. Springer, Cham, 2019.

POSTERS

Lin Yuyu, Kai Zheng, Lijuan Liu, Yang Chen, Jiahao Guo, Shuo Li, Cheng Yao, and Fangtian Ying. "OlfacEnhancer: A Vision-Based Scented Necklace for Cross-Modal Perception and Olfaction Augmentation." *Design and Semantics of Form and Movement*: 237.

Chen Yang, **Yuyu Lin**, Lijuan Liu, Cheng Yao, and Fangtian Ying. "Shadower: applying shadows to children's outdoor interaction." In *Adjunct Proceedings of the 2019 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2019 ACM International Symposium on Wearable Computers*, pp. 33-36. 2019.

Chen Yang, **Yuyu Lin**, Junwu Wang, Lijuan Liu, Cheng Yao, and Fangtian Ying. "IPANDA: a playful hybrid product for facilitating children's wildlife conservation education." In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1-6. 2019.

HONORS, AWARDS, and FELLOWSHIPS

2024 Center for Machine Learning and Health (CMLH) Fellowship

2021 Chiang Chen Overseas Graduate Fellowship, Chiang Chen Industrial Charity Foundation

2021 Outstanding Graduates, Zhejiang Province

2015, 2020 National Scholarship (top 0.1%), Chinese Ministry of Education

2018 Outstanding Graduation Design Award, Zhejiang University

2017 Outstanding Award of Scientific Research Training Program, Zhejiang University

2016 1st Prize of Outstanding Humanity & Social Science Research Achievement, Zhejiang University

2016 Morningside Cultural China Scholars

2015 2nd Prize of National Competition of "Challenge Cup" Academic Science and Technology, China Association for Science and Technology, Ministry of Education, All-China Federation of Students

TEACHING

2024 Interactive Data Science. Carnegie Mellon University.
Teaching assistant for Prof. John Stamper

2024 Programming user interface. Carnegie Mellon University.
Teaching assistant for Prof. Sauvik Das

2022 Introduction to Human-Computer Interaction. Stanford University.
Teaching assistant for Prof. James Landay

ACADEMIC SERVICE

Reviewer. UIST 2025; CHI 2023-2025; SIGGRAPH poster 2023-2025; CHI LBW AC 2024-2025; UIST poster 2025.

CMU GSAP 2023-2024

OurCS workshop 2024

Student volunteer. CHI 2023, 2025. UIST 2024 (paper chair assistant)

RESEARCH MENTORSHIP

09/2025-12/2025 **Emma Kim.** MSE & BME, CMU. Undergraduate Research Assistant
Olapeju Otusajo. MHCI, CMU. Master's Student Research Assistant

05/2025-08/2025 **Hailey Kim.** ME, CMU. Undergraduate Research Assistant
Yuchen Zhou. CS & HCI. CMU, Undergraduate Research Assistant
Anvita Nadgir. ECE. CMU, Undergraduate Research Assistant

09/2024-04/2025 **Dian Zhu.** Math. CMU, Undergraduate Research Assistant
Anoushka Naidu. CS. CMU, Undergraduate Research Assistant

05/2024-08/2024 **Deon Harper.** ME. Pennsylvania State University. Research Assistant through Research Experiences for Undergraduates (REU)
Kenneth Yu. Design. CMU, Undergraduate Research Assistant
Mingzhen Zhao. CS. CMU, Master's Student Research Assistant

09/2023-05/2024 **Sonia Prashant.** Architecture. CMU, Undergraduate Research Assistant